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# VIITJEE BY VENKATESH

Limits: 50 IIT JEE Objective Questions

## Limits · 50 Questions

Select answer → check & get solution

Question 1

$$\lim_{x \to 0} \frac{\sin 5x}{x}$$

- $0$
- $5$
- $1$
- Does not exist

Check Answer

✅ **Correct answer:** B. $5$

💡 **Solution:** Use $\lim_{u \to 0} \frac{\sin u}{u} = 1$. Here $\frac{\sin 5x}{x} = 5 \cdot \frac{\sin 5x}{5x} \to 5 \cdot 1 = 5$.

Question 2

$$\lim_{x \to 0} \frac{1 - \cos 2x}{x^2}$$

- $0$
- $1$
- $2$
- $4$

Check Answer

✅ **Correct answer:** C. $2$

💡 **Solution:** $1 - \cos 2x = 2\sin^2 x$. Then limit = $2 \lim_{x \to 0} \frac{\sin^2 x}{x^2} = 2 \cdot 1^2 = 2$.

Question 3

$$\lim_{x \to 0} \frac{\tan x - \sin x}{x^3}$$

- $\frac{1}{2}$
- $\frac{1}{3}$
- $1$
- $0$

Check Answer

✅ **Correct answer:** A. $\frac{1}{2}$

💡 **Solution:** Write $\tan x - \sin x = \sin x \frac{1 - \cos x}{\cos x}$. Using $\sin x \sim x$, $1 - \cos x \sim x^2/2$, gives $\frac{x \cdot x^2/2}{x^3 \cos x} \to \frac{1}{2}$.

Question 4

$$\lim_{x\to\infty}\left(1+\frac{2}{x}\right)^{3x}$$

- $e^2$
- $e^3$
- $e^6$
- $e$

Check Answer

✅ **Correct answer:** C. $e^6$

💡 **Solution:** $\left(1+\frac{2}{x}\right)^{3x} = \left[\left(1+\frac{2}{x}\right)^{x}\right]^3 \to (e^2)^3 = e^6$.

Question 5

$$\lim_{x\to 0}\frac{e^{2x}-1}{x}$$

- $0$
- $1$
- $2$
- $\infty$

Check Answer

✅ **Correct answer:** C. $2$

💡 **Solution:** Standard limit $\lim_{u\to 0}\frac{e^u-1}{u}=1$. Here $\frac{e^{2x}-1}{x} = 2\cdot\frac{e^{2x}-1}{2x} \to 2\cdot 1 = 2$.

Question 6

$$\lim_{x \to 0} \frac{\ln(1+3x)}{x}$$

- $1$
- $3$
- $\frac{1}{3}$
- $0$

Check Answer

✅ **Correct answer:** B. $3$

💡 **Solution:** $\lim_{u \to 0} \frac{\ln(1+u)}{u} = 1$. Then $\frac{\ln(1+3x)}{x} = 3 \cdot \frac{\ln(1+3x)}{3x} \to 3$.

Question 7

$$\lim_{x \to 0} \frac{7^x - 1}{x}$$

- $\ln 7$
- $7$
- $1$
- $0$

Check Answer

✅ **Correct answer:** A. $\ln 7$

💡 **Solution:** $\lim_{x \to 0} \frac{a^x - 1}{x} = \ln a$. So $\frac{7^x - 1}{x} \to \ln 7$.

Question 8

$$\lim_{x \to 0} \frac{x - \sin x}{x^3}$$

- $\frac{1}{3}$
- $\frac{1}{6}$
- $\frac{1}{2}$
- $0$

Check Answer

✅ **Correct answer:** B. $\frac{1}{6}$

💡 **Solution:** Use series: $\sin x = x - x^3/6 + \cdots$, so $x - \sin x \sim x^3/6$, giving $1/6$.

Question 9

$$\lim_{x \to 0} \frac{\tan 4x}{x}$$

- $0$
- $1$
- $4$
- $\infty$

Check Answer

✅ **Correct answer:** C. $4$

💡 **Solution:** $\frac{\tan 4x}{x} = 4 \cdot \frac{\tan 4x}{4x} \to 4 \cdot 1 = 4$.

Question 10

$$\lim_{x \to \pi/2} \frac{\cos x}{x - \pi/2}$$

- $1$
- $-1$
- $0$
- $\infty$

Check Answer

✅ **Correct answer:** B. $-1$

💡 **Solution:** Let $t = x - \pi/2$, then $\cos x = \cos(t + \pi/2) = -\sin t$. Limit becomes $\lim_{t \to 0} \frac{-\sin t}{t} = -1$.

Question 11

$$\lim_{x \to 0} \frac{\sqrt{1+x} - 1}{x}$$

- $0$
- $\frac{1}{2}$
- $1$
- $\frac{3}{2}$

Check Answer

✅ **Correct answer:** B. $\frac{1}{2}$

💡 **Solution:** Rationalise: $\frac{(\sqrt{1+x}-1)(\sqrt{1+x}+1)}{x(\sqrt{1+x}+1)} = \frac{x}{x(\sqrt{1+x}+1)} \to \frac{1}{2}$.

Question 12

$$\lim_{x \to 0} \frac{\sin^{-1} x}{x}$$

- $0$
- $1$
- $\infty$
- does not exist

Check Answer

✅ **Correct answer:** B. $1$

💡 **Solution:** Standard limit $\lim_{x \to 0} \frac{\arcsin x}{x} = 1$.

Question 13

$$\lim_{x \to 0} \frac{\tan^{-1} x}{x}$$

- $0$
- $1$
- $-1$
- $\frac{\pi}{4}$

Check Answer

✅ **Correct answer:** B. $1$

💡 **Solution:** Standard limit $\lim_{x \to 0} \frac{\arctan x}{x} = 1$.

Question 14

$$\lim_{x \to \infty} \frac{\sin x}{x}$$

- $1$
- $0$
- oscillates
- $\infty$

Check Answer

✅ **Correct answer:** B. $0$

💡 **Solution:** $|\sin x| \leq 1$ and $x \to \infty$, so by squeeze theorem limit = 0.

Question 15

$$\lim_{x \to 0} x \cot x$$

- $0$
- $1$
- $\infty$
- $\frac{\pi}{2}$

Check Answer

✅ **Correct answer:** B. $1$

💡 **Solution:** $x \cot x = \frac{x}{\tan x} \to 1$.

Question 16

$$\lim_{x \to 0} \frac{1 - \cos x}{x \sin x}$$

- $\frac{1}{2}$
- $1$
- $0$
- $2$

Check Answer

✅ **Correct answer:** A. $\frac{1}{2}$

💡 **Solution:** $1 - \cos x \sim x^2/2$, $x \sin x \sim x^2$, ratio $\to 1/2$.

Question 17

$$\lim_{x \to 0} \frac{e^x - e^{-x}}{x}$$

- $0$
- $1$
- $2$
- $\infty$

Check Answer

✅ **Correct answer:** C. $2$

💡 **Solution:** $e^x - e^{-x} = 2 \sinh x \sim 2x$, so limit = 2.

Question 18

$$\lim_{x \to a} \frac{x^n - a^n}{x - a}$$

- $na^{n-1}$
- $a^{n-1}$
- $na^n$
- $0$

Check Answer

✅ **Correct answer:** A. $na^{n-1}$

💡 **Solution:** Derivative definition: $\lim_{x \to a} \frac{x^n - a^n}{x - a} = na^{n-1}$.

Question 19

$$\lim_{x \to 1} \frac{x^m - 1}{x^n - 1}$$

- $\frac{m}{n}$
- $\frac{n}{m}$
- $1$
- $0$

Check Answer

✅ **Correct answer:** A. $\frac{m}{n}$

💡 **Solution:** Factor $x^k - 1 = (x - 1)(x^{k-1} + \cdots + 1)$, ratio $\to m/n$.

Question 20

$$\lim_{x \to 0} \frac{\sin 2x\,(1 - \cos 2x)}{x^3}$$

- $2$
- $4$
- $0$
- $8$

Check Answer

✅ **Correct answer:** B. $4$

💡 **Solution:** $\sin 2x \sim 2x$, $1 - \cos 2x \sim (2x)^2/2 = 2x^2$, product $\sim 4x^3$, limit=4.

Question 21

$$\lim_{x \to 0} \frac{\int_0^x \sin t^2\, dt}{x^3}$$

- $\frac{1}{3}$
- $\frac{1}{2}$
- $0$
- $1$

Check Answer

✅ **Correct answer:** A. $\frac{1}{3}$

💡 **Solution:** By Fundamental Theorem and L'Hôpital: numerator derivative $\sin x^2 \sim x^2$, so $\frac{x^2}{3x^2} \to \frac{1}{3}$.

Question 22

$$\lim_{x \to 0} \frac{\tan x - x}{x^3}$$

- $\frac{1}{2}$
- $\frac{1}{3}$
- $\frac{1}{6}$
- $0$

Check Answer

✅ **Correct answer:** B. $\frac{1}{3}$

💡 **Solution:** Series: $\tan x = x + x^3/3 + \cdots$, so $\tan x - x \sim x^3/3$, limit = 1/3.

Question 23

$$\lim_{x \to 0} \frac{x - \tan^{-1} x}{x^3}$$

- $\frac{1}{3}$
- $\frac{1}{2}$
- $1$
- $0$

Check Answer

✅ **Correct answer:** A. $\frac{1}{3}$

💡 **Solution:** Series: $\arctan x = x - x^3/3 + \cdots$, so $x - \arctan x \sim x^3/3$, limit=1/3.

Question 24

$$\lim_{x \to 0} \frac{\ln(1+x) - x}{x^2}$$

- $\frac{1}{2}$
- $-\frac{1}{2}$
- $1$
- $-1$

Check Answer

✅ **Correct answer:** B. $-\frac{1}{2}$

💡 **Solution:** Series: $\ln(1+x) = x - x^2/2 + \cdots$, so numerator $\sim -x^2/2$, limit = -1/2.

Question 25

$$\lim_{x \to 0} \frac{e^x - 1 - x}{x^2}$$

- $\frac{1}{2}$
- $0$
- $1$
- $\frac{1}{6}$

Check Answer

✅ **Correct answer:** A. $\frac{1}{2}$

💡 **Solution:** Series: $e^x = 1 + x + x^2/2 + \cdots$, so $e^x - 1 - x \sim x^2/2$, limit=1/2.

Question 26

$$\lim_{x \to 0} \frac{\sin x - \tan x}{x^3}$$

- $-\frac{1}{2}$
- $\frac{1}{2}$
- $-\frac{1}{3}$
- $\frac{1}{3}$

Check Answer

✅ **Correct answer:** A. $-\frac{1}{2}$

💡 **Solution:** Series: $\sin x = x - x^3/6$, $\tan x = x + x^3/3$, difference $\sim -x^3/2$, limit=-1/2.

Question 27

$$\lim_{x \to 0} (1 + \sin x)^{\csc x}$$

- $e$
- $1$
- $\infty$
- $0$

Check Answer

✅ **Correct answer:** A. $e$

💡 **Solution:** Let $y = \sin x$, then expression = $(1 + y)^{1/y} \to e$.

Question 28

$$\lim_{x \to 0} (1 + 3x)^{1/x}$$

- $e^3$
- $e$
- $3$
- $1$

Check Answer

✅ **Correct answer:** A. $e^3$

💡 **Solution:** Standard limit $\lim_{u \to 0} (1 + u)^{1/u} = e$. Here $(1 + 3x)^{1/x} = [(1 + 3x)^{1/(3x)}]^3 \to e^3$.

Question 29

$$\lim_{x \to \infty} \left( \frac{x + 2}{x - 1} \right)^x$$

- $e^3$
- $e^2$
- $e^{-3}$
- $e$

Check Answer

✅ **Correct answer:** A. $e^3$

💡 **Solution:** $\frac{x+2}{x-1} = 1 + \frac{3}{x-1}$, limit = $e^{\lim x \cdot \frac{3}{x-1}} = e^3$.

Question 30

$$\lim_{x \to 0} \frac{\cos x - e^{-x^2/2}}{x^4}$$

- $-\frac{1}{12}$
- $\frac{1}{12}$
- $0$
- $-\frac{1}{6}$

Check Answer

✅ **Correct answer:** A. $-\frac{1}{12}$

💡 **Solution:** Series: $\cos x = 1 - x^2/2 + x^4/24$, $e^{-x^2/2} = 1 - x^2/2 + x^4/8$, difference $\sim -x^4/12$, limit=-1/12.

Question 31

$$\lim_{x \to 0} \frac{\sin(x^2)}{x^2}$$

- $0$
- $1$
- $\infty$
- does not exist

Check Answer

✅ **Correct answer:** B. $1$

💡 **Solution:** Let $u = x^2 \to 0$, then $\frac{\sin u}{u} \to 1$.

Question 32

$$\lim_{x\to 0} \frac{\log_{\sin x} \cos x}{\log_{\sin(x/2)} \cos(x/2)}$$

- $2$
- $4$
- $1$
- $8$

Check Answer

✅ **Correct answer:** B. $4$
💡 **Solution:** Use change of base and approximations: $\cos x \sim 1 - x^2/2$, $\sin x \sim x$. After simplification limit = 4.

Question 33

$$\lim_{x\to 0} \frac{x^2 \sin(1/x)}{\sin x}$$

- $0$
- $1$
- $\infty$
- does not exist

Check Answer

✅ **Correct answer:** A. $0$
💡 **Solution:** $|x^2 \sin(1/x)| \leq x^2$ and $\sin x \sim x$, so expression $\sim x \sin(1/x) \to 0$.

Question 34

$$\lim_{x\to\infty} \frac{\ln x}{x}$$

- $0$
- $1$
- $\infty$
- $e$

Check Answer

✅ **Correct answer:** A. $0$

💡 **Solution:** Logarithm grows slower than any positive power, so limit = 0.

Question 35

$$\lim_{x\to 0} \left(\frac{\sin x}{x}\right)^{1/x^2}$$

- $e^{-1/6}$
- $e^{1/6}$
- $1$
- $e^{-1/3}$

Check Answer

✅ **Correct answer:** A. $e^{-1/6}$

💡 **Solution:** Take log: $\frac{1}{x^2}\ln\left(\frac{\sin x}{x}\right) \sim \frac{1}{x^2}\cdot(-x^2/6) \to -1/6$, so limit = $e^{-1/6}$.

Question 36

$$\lim_{x \to 0} \frac{e^{x^2} - \cos x}{x^2}$$

- $\frac{3}{2}$
- $1$
- $\frac{1}{2}$
- $2$

Check Answer

✅ **Correct answer:** A. $\frac{3}{2}$

💡 **Solution:** Series: $e^{x^2} = 1 + x^2 + \cdots$, $\cos x = 1 - x^2/2 + \cdots$, difference $\sim (3/2)x^2$, limit=3/2.

Question 37

$$\lim_{x \to 0} \frac{\sin x - x}{x^3}$$

- $-\frac{1}{6}$
- $\frac{1}{6}$
- $-\frac{1}{3}$
- $0$

Check Answer

✅ **Correct answer:** A. $-\frac{1}{6}$

💡 **Solution:** Series: $\sin x = x - x^3/6 + \cdots$, so $\sin x - x \sim -x^3/6$, limit = -1/6.

Question 38

$$\lim_{x \to 0} \frac{\tan 2x - 2x}{x^3}$$

- $\frac{8}{3}$
- $\frac{4}{3}$
- $\frac{2}{3}$
- $0$

Check Answer

✅ **Correct answer:** A. $\frac{8}{3}$

💡 **Solution:** Series: $\tan u = u + u^3/3 + \cdots$, with $u = 2x$ gives $2x + 8x^3/3 - 2x = 8x^3/3$, limit = 8/3.

Question 39

$$\lim_{x \to 0} \frac{\ln(1 + x^2)}{x \sin x}$$

- $1$
- $0$
- $2$
- $\infty$

Check Answer

✅ **Correct answer:** A. $1$

💡 **Solution:** $\ln(1 + x^2) \sim x^2$, $x \sin x \sim x^2$, ratio $\to 1$.

Question 40

$$\lim_{x\to 0} \frac{\sin 3x}{\sin 7x}$$

- $\frac{3}{7}$
- $\frac{7}{3}$
- $1$
- $0$

Check Answer

✅ **Correct answer:** A. $\frac{3}{7}$

💡 **Solution:** $\frac{\sin 3x}{\sin 7x} = \frac{3x\cdot(\sin 3x/(3x))}{7x\cdot(\sin 7x/(7x))} \to \frac{3}{7}$.

Question 41

$$\lim_{x\to 0} \frac{xe^x - \ln(1+x)}{x^2}$$

- $\frac{3}{2}$
- $1$
- $\frac{1}{2}$
- $2$

Check Answer

✅ **Correct answer:** A. $\frac{3}{2}$

💡 **Solution:** Series: $xe^x = x(1+x+\cdots) = x + x^2 + \cdots$, $\ln(1+x) = x - x^2/2 + \cdots$, difference $\sim (3/2)x^2$, limit=3/2.

Question 42

$$\lim_{x \to \pi/4} \frac{\sin x - \cos x}{x - \pi/4}$$

- $\sqrt{2}$
- $1$
- $0$
- $-\sqrt{2}$

Check Answer

✅ **Correct answer:** A. $\sqrt{2}$

💡 **Solution:** Derivative of $\sin x - \cos x$ at $x = \pi/4$ is $\cos x + \sin x = \sqrt{2}$.

Question 43

$$\lim_{x \to 0} \frac{\sqrt{1 + x \sin x} - \sqrt{\cos x}}{x \tan x}$$

- $\frac{3}{4}$
- $\frac{1}{2}$
- $\frac{1}{4}$
- $1$

Check Answer

✅ **Correct answer:** A. $\frac{3}{4}$

💡 **Solution:** Rationalise numerator, use approximations $\sqrt{1+u} \approx 1 + u/2$, after simplification limit = 3/4.

Question 44

$$\lim_{x\to 0} \frac{(1+x)^{1/x} - e}{x}$$

- $-\frac{e}{2}$
- $\frac{e}{2}$
- $-e$
- $e$

Check Answer

✅ **Correct answer:** A. $-\frac{e}{2}$

💡 **Solution:** Expand $(1+x)^{1/x} = e^{1-x/2+\cdots}$, so numerator $\sim -ex/2$, limit = -e/2.

Question 45

$$\lim_{x\to 0} \frac{\cos(\sin x) - \cos x}{x^4}$$

- $\frac{1}{6}$
- $\frac{1}{12}$
- $\frac{1}{3}$
- $0$

Check Answer

✅ **Correct answer:** A. $\frac{1}{6}$

💡 **Solution:** Use $\cos u - \cos v = -2\sin\frac{u+v}{2}\sin\frac{u-v}{2}$. Approx $\sin x \approx x - x^3/6$, leads to limit = 1/6.

Question 46

$$\lim_{x\to 0} \frac{\int_0^{x^2} \sin t^2 \, dt}{x^6}$$

- $\frac{1}{3}$
- $\frac{1}{6}$
- $0$
- $\frac{1}{2}$

Check Answer

✅ **Correct answer:** A. $\frac{1}{3}$

💡 **Solution:** Numerator derivative: $2x \sin(x^4) \sim 2x^5$, integrate $\sim 2x^6/6 = x^6/3$, limit = 1/3.

Question 47

$$\lim_{x\to 0} \frac{\sin(\tan x) - \tan(\sin x)}{x^7}$$

- $-\frac{1}{30}$
- $\frac{1}{30}$
- $0$
- $-\frac{1}{15}$

Check Answer

✅ **Correct answer:** A. $-\frac{1}{30}$

💡 **Solution:** Series expansion: $\sin(\tan x) - \tan(\sin x) \sim -x^7/30$, limit = -1/30.

Question 48

$$\lim_{x\to\infty}\left(\frac{x^2+5x+3}{x^2+x+2}\right)^x$$

- $e^4$
- $e^3$
- $e^2$
- $e$

Check Answer

✅ **Correct answer:** A. $e^4$

💡 **Solution:** Ratio = $1+\frac{4x+1}{x^2+x+2}$, exponent $x\cdot\frac{4x+1}{x^2+x+2}\to 4$, limit = e^4.

Question 49

$$\lim_{x\to 0}\frac{\cos x-\cos 3x}{x^2}$$

- $4$
- $2$
- $0$
- $8$

Check Answer

✅ **Correct answer:** A. $4$

💡 **Solution:** Use identity: $\cos x-\cos 3x=2\sin 2x\sin x\sim 2\cdot(2x)\cdot x=4x^2$, limit=4.

Question 50

$$\lim_{x \to 0} \frac{2^x - 3^x}{x}$$

- $\ln \frac{2}{3}$
- $\ln \frac{3}{2}$
- $0$
- $1$

Check Answer

✅ **Correct answer:** A. $\ln \frac{2}{3}$

💡 **Solution:** Write $2^x - 3^x = 3^x((2/3)^x - 1)$, then $\frac{3^x((2/3)^x - 1)}{x} \to \ln(2/3)$.

Tap "Check Answer" for individual feedback, or "See All Solutions" to reveal all answers at once.